DIFF-REF-STRUCT WEB

User manual

Powder diffraction series in the browser: every control, what the core computes behind it, the limits of this version

For anyone who opens a folder of diffraction patterns in the page.

For diff-ref-struct Web 1.0.0 with core 0.81.0 · 2026-10-27

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jmigartua.github.io/diff-ref-struct-web (the app) · …/guide (this manual online)

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For **diff-ref-struct Web 1.0.0 with core 0.81.0**, at https://jmigartua.github.io/diff-ref-struct-web/. The page shows the version it runs at the foot of its start page and in the line under the inspector, and its **Help** ▾ menu opens the manual of that version. The manual ends with three appendices made from the guide's records: **A, Conventions** (the words, files, suffixes and names), **B, Release notes** (every version up to this one) and **C, Roadmap**.

The site's **Recipes** work through the analyses the page is made for (a temperature ramp, a hold at constant temperature, two patterns compared, a run followed live), and its **Quick reference** fits the controls on one page.

1. What the page is

diff-ref-struct Web is the **Plots module** of the diff-ref-struct Mac app, running in a browser page. It shows many powder-diffraction patterns at once as a **plot set**. You can view them as:

- an overlay;
- a stack;
- a WinPLOTR-style projection;
- a colour **map** of intensity against 2θ and the set's variable (temperature or time);
- a difference map against a reference pattern;
- a 3D surface;
- a series of quantities taken from windows of 2θ.

Three things set it apart from a plotting web page:

- **The numbers are the core's.** The page contains the diff-ref-struct core, the same Rust engine the Mac app uses, compiled to WebAssembly. The core computes every number on screen: reading the files, the temperatures found in their names, the map's grid and colour range, the stack's steps, the projection, the differences, the windows' integrals. The page draws what the core returns and computes nothing itself. A map in the browser therefore equals the map in the app, number for number.
- **Your files never leave your machine.** The browser reads a folder or files where they are. Nothing is uploaded, there is no server behind the page, and you need no account. The page has no tracking. The line at the foot of the inspector says so as well.
- **It recognises series.** Open a folder of scans from a temperature ramp or a time-resolved run and the core proposes the plot sets it finds: one scan per temperature, each hold at constant temperature against time, and the whole run against time. Each proposal is one click away.

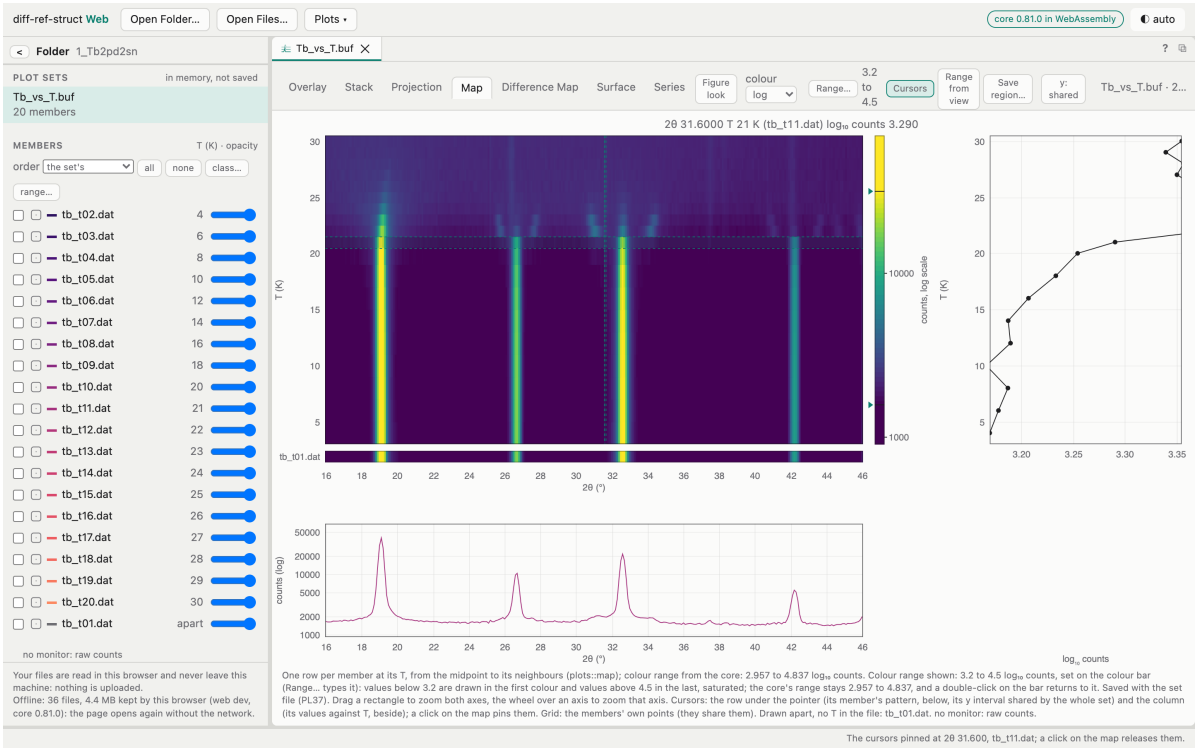
What it is **not**: it is not the refinement program. The Rietveld refinement, the Data module's operations, Publish's figures and the project's files belong to the Mac app. The browser page is for looking at data, comparing patterns, following a series and finding where things change.

2. What you need

what	needed
Browser	Any current browser. **Chromium-based browsers** (Chrome, Edge, Arc, Brave) do the most: they open a folder in place, so that **live mode** works (§15). Firefox and Safari open a folder through the browser's folder dialog instead, which takes a copy of the list once; live mode is not offered there.
Network	Only for the first visit. After it the page keeps itself and opens offline (§17).
Memory	Ordinary sets (tens of files) need nothing special. A run of a hundred synchrotron scans of 27 000 points each (112 MB) works; allow a few hundred MB of memory for the tab.
Files	Text data files up to **64 MB** each (§21). Larger files are listed but not read.

Permissions. When you open a folder, Chromium asks "Let site view files?". The page asks only to *read* and never writes into the folder. Firefox and Safari may word their dialog as an "upload". It is not one: the files stay in the browser.

3. The page at a glance



The page with a plot set open: the Tb_2Pd_2Sn patterns of FullProf's WinPLOTR tutorial on the Map, colour on a log scale clipped to 3.2–4.5, the cursors pinned at 2θ 31.6° (row strip below, column strip at the right). At the left, the inspector with the plot set and its members.

From top to bottom: **the bar**; at the left **the inspector**, and the **separator** at its right edge; at the right **the views**, one tab per plot set, each with its view tabs, toolbar, plot and note; at the foot **the status**

line.

The bar, left to right:

- **Open Folder…**: chooses a folder of data files (§4).
- **Open Files…**: chooses one or more files: data files, a plot set file (`.drv`) or a WinPLOTR buffer (`.buf`).
- **Plots ▾**: the module's menu, the same list as the app's Plots menu (§14). Items that cannot be used now are greyed out, and hovering one says why.
- **The core's chip**: "core 0.81.0 in WebAssembly" once the core has loaded. If it reads "core: waiting…", the core did not load; reload the page.
- **Help ▾**: the guide; **Manual for this view** (the manual opened at the section of the view shown); **What's new in this version**; **Report a problem…** and **Ask for a feature…** (a form of the public repository's issues, filled in with the version, the browser and the view's address, never your data; or the same by e-mail); **Cite this view** (one line for a lab book or a paper's methods, copied to the clipboard).
- **Appearance**: "◐ auto" (the system's), "☀ light", "☾ dark". Click for the next one. The choice is kept.

The inspector (the column at the left), top to bottom:

- **Folder**: its name. < shrinks the inspector to a narrow rail and > brings it back.
- **Files**: the folder's tree (§5), with **Open as Plot Set** and, in Chromium, the **Live** controls (§15).
- **Sets in this folder**: the core's proposals (§7).
- **Plot sets, Windows, Regions, Members**: the panels of the plot set shown (§9).
- **The member**: the facts of the member chosen (fold it open).
- **The privacy line**, and under it the offline line (§17).

The separator between inspector and views: drag it to set the inspector's width (220–640 px), or double-click it to shrink the inspector to a rail and back. With the keyboard, focus it and use ← → (16 px a step, 64 px with ⇧). The width is kept for your next visit. On a narrow window (under 640 px) the inspector stacks above the views.

The views take the rest of the page: one tab per plot set, each with its view tabs and toolbar (§8). Before anything is open, a start page shows the two Open buttons and the version.

The status line at the foot says what the last action did ("Plots: 39 members …") or why it could not. Messages in red are refusals or problems. Read it whenever something seems not to happen.

4. Bringing files in

There are three ways, and every one of them reads the files in the browser only. A fourth needs no files of yours: **Try the example** on the start page opens a synthetic heating run computed by the core (21 scans of a periclase sample from 300 to 1300 K, with corundum appearing above about 1000 K), as if its folder had been dropped.

Open Folder… (bar or start page). Choose a folder. Its files appear in the tree, sub-folders included. Hidden files and folders (names starting with `.`, such as `.DS_Store` or `.git`) are left out. The page lists the files at once and reads a file's content only when the core needs it. Then:

1. The core **describes** every file, one after another: what it is (an `.xye` file, a FullProf free-format file, a buffer, a plot set file …, or "?" for something no reader recognises). The tree fills in each file's kind as the answers come. For a hundred large synchrotron files this takes about a minute.
2. The core **proposes** the plot sets the folder's data files make (§7).
3. In Chromium, **live mode** starts watching the folder (§15).

Open Files… (bar or start page). Choose one or more files. They join the folder already shown, or make a new one if none is, and they **open in Plots at once** (§6 says as what).

Drag and drop. Drop anything from the Finder (or Explorer) anywhere on the page:

- **a folder** becomes the folder shown, exactly as with Open Folder…. In Chromium a dropped folder is also watched by live mode;
- **loose files** join the folder shown and open in Plots at once, as with Open Files…. Dropping the same files again opens them again.

A page holds **one folder at a time**. Opening another folder replaces the tree. Plot sets already open stay in their tabs.

5. The folder's tree

Each row of the tree shows a file's name, its kind as the core says it (`xye`, `free_steps`, `buffer_file`, `plot_set`, or `?`) and its size. Hover a row for the file's path, its size and the core's full description.

folders | kinds (the switch on the Files header) chooses how the tree is listed:

- **folders**: as on disk. The first level of sub-folders is open, and a double-click (or click) on a folder opens or closes it.
- **kinds**: one group per kind as the core describes the files: the data files first, then the project's files (plot sets, buffers), then what is not recognised. This makes it easy to see at a glance which files the core can read.

The choice is kept for your next visit. The header also counts the files and their total size.

Choosing files. A click chooses a file, **⌘-click** (Ctrl-click on Windows and Linux) adds or removes one, and **⇧-click** chooses a range. **Open as Plot Set** then opens the chosen files; it says "Open 12 as Plot Set" when 12 are chosen.

Opening directly.

- A **double-click** on a file opens it in Plots.
- The **set** button on a folder's row opens all the data files under that folder as one plot set. This is the quickest way when a folder holds exactly one series.

6. What opens as what

Whatever you open (a double-click, Open as Plot Set, a folder's "set", files dropped or picked), the page asks the core what each file is and routes it accordingly:

What is among the files	What opens
a **plot set file** (`.drv`)	the set it describes, with its members, variables, view settings, clips and regions. Its member files are taken from the folder by the names the `.drv` keeps (see below).
a **WinPLOTR buffer** (`.buf`)	the buffer's set, its files taken from the folder by the names it lists.
data files only	one new plot set with those files as its members, in name order. The core reads each member's variable from the file if it can: a temperature in its header, or in its name (`_300K`, `T300`, `T0900C` …), or an elapsed time (`t16530s`).
something else (a `.pcr`, an image, an unrecognised file)	left out. The status line names it and says what the core says it is.

A `.drv` or `.buf` needs its files where it names them. A plot set file written by the app names its members relative to the project folder, for example `Data/scan_001.xye` for a `.drv` kept in `Plots/`. So open the **project folder** (the one that holds `Plots/` and `Data/`), not the `Plots/` folder alone, and then open the `.drv` from the tree. A member whose file is missing is named in the status line.

The same set file opened twice brings its tab forward instead of opening a second copy.

7. Sets in this folder: the core's proposals

When a folder is opened, the core reads all its data files and proposes the plot sets they make. The section **Sets in this folder** lists them, each with an **Open** button, a title and the number of members. Hover a proposal for the core's kind and exposure class and for the files it leaves out, with the reason.

The kinds of proposal:

- **By temperature**: one scan per distinct temperature, ordered. Where several scans share a temperature, the one of the longest exposure is taken. Examples: *"19 scans by temperature, 4 to 30 K"* (the Tb_2Pd_2Sn neutron patterns of FullProf's WinPLOTR tutorial); *"25 scans by temperature, 900 to 1300 °C (the longest exposure at each temperature, of 3 exposure classes), rebinned to 0.01°"* (an in-situ synchrotron run of 101 scans).
- **Against time**: the whole run, each scan at its time since the first. Example: *"100 scans against time, 10.036 h from …001… (3 exposure classes), rebinned to 0.01°"*.
- **A hold**: a run of three or more scans at the same temperature, against time since the hold's first scan. Other holds of the same class are named in the title. Example: *"the hold at 1270 °C: 39 scans against time, 166.433 min …"*.

Where the variable comes from:

- **The temperature** is taken from a file's header or its name.
- **The time** is taken from the name (`t00000s`) or, failing that, from the files' modification times.
- **Exposure classes** separate scans counted differently, such as long and short scans. The core finds them from the counts' scale, the median of σ^2/y above 2°. Scans of one class differ in that scale by less than 10 %. When there is more than one class, the section says how many.

Rebinning on opening. For large scans (tens of thousands of points), the proposal says *"rebinned to 0.01°"*. The core then averages the points onto a 0.01° grid as the set opens, so that the map and the overlay draw in about a second instead of minutes. The files themselves are not touched.

Files that fit no proposal (a background scan with no temperature in its name, for example) are listed under "Left out of every set" with the reason.

Open opens the proposal as a new plot set. A proposal opened this way is also remembered for **live mode**: when new scans of the same kind and class appear in the folder, they are added to it (§15).

> **Important in this version: proposals against time.** A proposal opens from its files, and each member's variable is read again from the file, not from the proposal. Where a file's name gives both a temperature and a time (as in `…_T1270C_t16530s_…`), the members open **by temperature**, with the time only as a second variable. A **by-temperature** proposal is therefore right. A **hold** or a **run against time** opens with every scan of a hold at the same temperature, so its map collapses onto one line. Until this is fixed, open sets against time from a plot set file (`.drv`) that types each scan's time (the site's recipe *A hold: growth curves* shows how). The status line says when a proposal's values were not carried.

8. A plot set in its tab

Each plot set opens in a tab of its own in the views area. Tabs can be dragged to rearrange them, and closed with their ×. The **?** at the top right of the views opens this manual at the section of the view shown. Above the plot, the tab shows:

- **The title strip**: the set's title, its number of members, and "· normalised" or "· in d" when a view rule is in force (§12).
- **The view tabs**: Overlay, Stack, Projection, Map, Difference Map, Surface, Series (§10).
- **The toolbar** of the view shown. Only the controls that apply to the view are visible:

control	in	does
y linear / √ / log	Overlay, Stack, Projection, Series	the intensity axis's scale
Linked zoom	Overlay	a second, zoomed plot beside the first, with a difference strip (§10)
Δ and its rule	Stack, Projection	the steps between members (Set Displacement…)
Draw window	Overlay, Stack	the next drag across the plot draws a window of 2θ (§12)
Figure look	all but Surface	the view as a figure: white, open axes, ink text, a scale bar for a stack or a projection

control	in	does
colour linear / √ / log	Map, Difference Map, Surface	the colour scale
Range… and its clip	Map, Difference Map, Surface	type the colour range (§11)
Cursors	Map, Difference Map	the row and column strips (§11)
Range from view	Map, Difference Map	the colour range follows the zoom (§11)
Save region…	Map, Difference Map	name the rectangle shown (§11)
y: shared / **y: own**	Overlay, Stack, maps	one y range for the whole set, or each member's own
of the map / the difference map	Surface	which map the surface shows
quantity, **background**, **+ window**	Series	what the series follows (§12)

Below the plot, a short note explains what the view shows and mentions anything left out or drawn apart.

9. The members and the member

When a plot set is shown, the inspector holds its panels:

Plot sets ("in memory, not saved"): one row per open set. Click a row to switch to that set.

Windows (2θ, °): the set's windows of 2θ (§12). Click a row to choose the window, double-click to rename it, and × removes it.

Regions: the named rectangles of the map (§11).

Members: one row per member, with its variable's name and unit in the header. Each row shows:

- a **checkbox**: whether the member is *chosen*. Chosen members are what Solo, Dim, Set Reference, Difference of Members and the inspector act on;
- an **icon** for the file's kind (a letter, for example F for FullProf formats, ≡ for a column table);
- a **colour swatch**: its colour in the overlay, along a dark-to-bright ramp in the set's order. Grey means a member with no value of the variable;
- the **label** (the file's name), with a badge "ops N" if the Mac app's Data module keeps operations on the file;
- the **value of the variable**, marked "typed" if it was typed rather than read, "apart" if it has none, or "out" if it is left out of the view;
- an **opacity** slider (0 to 1). Turn a member down to see others behind it.

Mouse on the member rows:

- a **click** chooses one member; **⌘-click** or **⇧-click** adds or removes a member;
- **⌥-click** (Alt-click) *solos* the member: the others are dimmed until you ⌥-click again;
- a **double-click** on a member without a value opens **Set Temperature…**;
- **hovering** shows the member's facts: file, value and its source, format, points, range, wavelength,

counting time and exposure class, monitor, the file's date, and the normalisation factor.

The control row above the members:

- **order**: the set's own order, by the variable, by the file's time, or by name;
- **all**, **none**: choose every member, or none;
- **class…**: choose the members of one exposure class, as the files state their counting time ("10 s: 4 members");
- **range…**: choose the members whose variable lies between two values, ends included.

Notes at the foot of the panel say, for example, "no monitor: raw counts", or which members are "Drawn apart (no T in the file)".

The member (fold it open): with nothing chosen it describes the set: whether all members share one grid, the counts, and how the view is set. With a member chosen it shows:

- the variable's value with its source, and other candidate values ("also: …");
- the Mac app's Data operations on the file;
- how the member is drawn;
- **What the file says**: format, axis, points, range, step, wavelength, monitor, counting time, title;
- the **Actions**: Set Temperature…, Solo Member, Set Reference, Set Wavelength… (and Open in Data, which belongs to the Mac app and does nothing in the browser).

Every value keeps its source. Set Temperature… asks for the value **and where it comes from** (a logbook, a thermocouple file, "nominal set point"). The page refuses a value without a source: the source is saved with the value. **Read Variable From…** reads the values again from the files' names or from the files themselves.

10. The views

All views share the same mouse controls (§18). In short: drag a box to zoom, double-click to see everything, ⇧-drag to pan.

Overlay

Every member drawn as a curve on the same axes, coloured along the set's order. Hover a curve to thicken it. ⌥-click a curve to solo it, and ⌥-click again to show all.

With **y: own**, the y range fits the chosen members. With **y: shared**, it is the core's extent of the whole set, so that the scale does not jump as members are switched.

Linked zoom splits the overlay in two:

- The left plot shows the whole pattern. Drag across it to choose the 2θ window that the right plot shows.
- Zooming or panning the right plot moves the band on the left.
- Under both plots a **difference strip** shows the first chosen member less the reference member, or less the second chosen member. It is the quickest way to see what changed between two

scans.

Large sets are drawn in slices. The status line counts them (“Overlay: drawing 60 of 100 members…”) while the page stays responsive.

Stack

Each member raised by a step above the one before and labelled at its baseline with its value (“1270 °C”), or with its name and “(no T)”. The core proposes the step from the data; **Δ** or **Set Displacement…** sets your own:

- **rule**: *step* (the same Δy for each member) or *variable* (Δy proportional to the variable, so that the spacing shows the temperature or time steps);
- **Δy**: the vertical step, in the y axis’s units;
- **Δx** (°): a horizontal shift per member, which gives a slanted stack. 0 gives a plain stack.

The y label reads “counts + each member’s step”: the numbers on the y axis hold for the first member only. In Figure look a scale bar replaces the y axis for that reason.

Projection

WinPLOTR’s pseudo-3D view: each member drawn behind the one before, shifted by Δx and Δy, its hidden parts removed, with an oblique axis carrying the variable. The view stays empty until you give it steps with **Δ** (Set the Projection’s Steps); it keeps steps of its own, separate from the Stack’s.

Map

Intensity as colour, with 2θ across and one row per member up the side at its value of the variable. Each row spans halfway to its neighbours, so uneven steps in temperature or time show as rows of different heights. This is the view for following peaks through a series: a phase appearing, a peak moving, a transition. §11 covers it in detail.

Difference Map

The map of each member **less a reference member**. Choose a member and **Set Reference** first (until you do, the warmest member placed is the reference). Values above zero and below zero are told apart by the colour ramp. The log scale is not offered, since differences can be negative: log falls back to signed √.

Surface

The map, or the difference map (**of** on the toolbar), as a 3D surface:

- drag to turn it;
- the wheel zooms;

- a double-click starts again;
- the pointer reads the cell under it.

The map's colour range also limits the height. **Copy Surface as Image** (menu) copies it to the clipboard as a PNG.

Series

The quantity chosen for each **window** of 2θ, against the variable, with ±σ error bars. Points below 2σ are drawn open. Each window has its own colour and marker. With no window yet, the view tells you to add one (§12).

11. The Map and the Difference Map in detail

Reading the map. The colour bar at the right gives the scale: "counts", "√counts" or "$\log_{10}$ counts". In the Difference Map it says what was subtracted ("less 1300 °C (name), counts, signed √ scale"). The readout follows the pointer and shows:

- 2θ;
- the variable;
- the member's label;
- any second variable of that row, with its source;
- the cell's value ("no value" where a cell is empty).

The right-hand axis. When the members carry a second variable (for a set against time, the temperature of each scan), the right-hand axis labels each row with it. You can then read the time on the left and the temperature on the right.

Colour scale (**colour** on the toolbar): linear, √ or log. √ is often best for diffraction: it compresses the strong peaks less than log and still shows the weak ones.

The colour range: the clip. The core gives the full range of the data. Strong peaks usually take the whole colour ramp, leaving the weak features in one colour. To go down to the details, **clip** the range:

- **Drag** either of the two marks on the colour bar (the pointer changes to a double arrow). Values beyond the clip are drawn saturated. The two marks stay in order and inside the core's range.
- **Range…** (or Plots ▸ Set Colour Range…) to **type** the two values, *From* and *To*. An empty field keeps the core's end at that side, and both empty restores the core's range. If the values lie wholly outside the data, the page refuses them and says why. If they hide most of the map, the core's warning is added to the note under the map ("The core says: …").
- **Double-click the colour bar** (or Reset Colour Range) to return to the core's range.

The clip is kept per view and per scale. It is saved with the set (§13) and carried in the page's address (§16). Changing the scale starts again from the core's range for that scale.

Range from view. Switched on, the colour range follows the zoom: whenever you zoom, the core gives the range of the rectangle shown, and the colours spread over that rectangle alone. Switched

off, the earlier clip comes back.

Zoom.

- **Drag a rectangle** to zoom both axes.
- **The wheel over the left margin** zooms the variable's axis only, and **over the bottom band** zooms 2θ only.
- **⌘-wheel** or a pinch zooms about the pointer.
- **A double-click** on the map shows the whole map.

The zoom is kept when you switch views and back.

Cursors (the toolbar's Cursors, or Row and Column Cursors):

- the **row strip** under the map shows the pattern of the row under the pointer, on the same 2θ axis as the map. In the Difference Map it shows the difference row;
- the **column strip** beside the map shows the intensity at the pointer's 2θ against the variable: a peak's growth through the series as a curve;
- **click the map** to **pin** the cursors at a point ("The cursors pinned at …"), then move freely; click again to release them.

Turning Cursors on switches to the Map if another view is shown. The strips' y range follows **y: shared / own** as in the Overlay. Cursors are not drawn in Figure look.

Regions: named rectangles. A region keeps a place of the map worth coming back to, such as a peak's onset or a window where a phase grows.

1. Zoom to the rectangle (and, if you like, switch Range from view on, or clip the colours).
2. **Save region…** and give it a name (r1, r2 … by default). The core computes the rectangle's block of rows and columns, and the status line names it. With Range from view on, the region keeps its colour range too.
3. The region is listed under **Regions** in the inspector. **Click** it to come back to it: the map zooms to the region and applies its own colour range.
4. **Double-click** a region (or Rename Region…) to rename it, and × (or Remove Region) to remove it.

Two regions cannot share a name. Regions are saved with the set (§13). A set file made in the Mac app brings its regions with it.

Members drawn apart. A member with no value of the variable has no place on the map's axis. It is drawn as a separate row and named in the note ("Drawn apart, no T in the file: …"). Give it a value with Set Temperature… to place it.

12. Normalising, converting x, windows and the Series view

These change **the view**, never the files ("A view: the files untouched", says the status line).

Normalise… puts the members on a common scale. It offers:

- **none**: the counts as read;

- **the monitor**: each member divided by the monitor its file states;
- **the counting time**: each member divided by the counting time its file states;
- **the maximum**: the highest point of each member set to the same value;
- **the same area**: each member's area set to the same value, between *From* and *To* 2θ;
- **a peak's integral**: one peak's integral above its background, between *From* and *To*;
- **a reference member**: every member scaled to the reference (choose it with Set Reference first).

The fields are *Normalised to* (the value), *From, 2θ (°)*, *To, 2θ (°)* and *Background* (auto, or a line between the ends). Members the rule cannot normalise (no monitor, for example) are left out of the view and listed. The title strip then shows "· normalised".

Convert X… draws the set against another variable:

- the files' own x;
- 2θ (°);
- d (Å);
- Q = 4π sin θ/λ (Å^{-1});
- sin θ/λ (Å^{-1});
- s = 1/d (Å^{-1}).

Every conversion but the first needs each member's wavelength. If some files state none, a chooser offers *a λ typed now* (with its source) or *none: leave them out*, listed. **Set Wavelength…** sets the λ of the chosen members, with its source.

Windows are ranges of 2θ followed through the set: a peak, a group of peaks, a band. There are three ways to make one:

- **+ window** on the Series toolbar, or **Add Window…**: a *Name*, *From (°)* and *To (°)*;
- **Draw window** (Overlay, Stack), then drag across the plot. The window is named w1, w2 …;
- the windows a `.drv` already carries.

To work with a window:

- drag a window's band edges on the plot to move them;
- click the band to choose the window;
- rename or remove it in the inspector or the menu.

The Series view plots, for every window, one **quantity** against the variable:

- **integral above the background** (counts·°), with σ;
- **centroid** (°), with σ;
- **strongest peak: position**, **height**, **width**;
- **background at x**: the background's value at the 2θ positions given with Background at….

The **background** under a window is either **automatic** (the arPLS baseline GSAS-II uses) or a **line between the ends** of the window. The integral of a peak's window against temperature or time is the usual way to follow a phase's growth.

13. Saving and exporting

Save Plot Set (Plots ▾) writes the set as a **plot set file**, `<name>.drv`. In the browser it is **downloaded** to your Downloads folder: nothing is written into the folder you opened. The `.drv` keeps:

- the members (by their file names) and the variable, with every typed value's source;
- the order;
- the stack's and projection's steps;
- the y and colour scales;
- the reference member;
- the windows, the opacities, the clips and the regions;
- the y interval.

It is a plain text (TOML) file, read by the Mac app and the command line `drs` as well as by this page.

To open it again later:

1. Move the downloaded `.drv` into the folder's `Plots/` sub-folder, or next to the data files. It must be placed where its member names resolve: §6.
2. Open the folder.
3. Double-click the `.drv`.

Or drop the `.drv` on the page while the folder is open.

Save Plot Set As…: the same, with a new name.

Copying. The bottom table's **Copy** (or ⌘C on selected rows) copies its rows as tab-separated text, ready to paste into a spreadsheet. **Copy Surface as Image** copies the surface as a PNG.

Viewer Page… (Dev): downloads a single HTML file, `<set> map viewer.html`, holding the map viewer and the set's maps, members, clips and regions. It opens in any browser, offline, without this page or the files. It is marked Dev because it is a developer's tool; it has not been checked in the browser host in this version.

Not in the browser in this version (they belong to the Mac app or need a later core build; §20):

- figure export (PDF, SVG, PNG);
- Plot Set as CSV…;
- Send to Publish….

A screenshot of the view in Figure look is the stop-gap for a figure.

14. The Plots menu, item by item

The **Plots** ▾ menu lists the module's commands in the app's order. A greyed item explains itself on hover. Marks: ✓ a toggle that is on; **(app)** an item that belongs to the Mac app and does nothing useful in the browser; **(later)** an item this browser core does not serve yet: it answers in the status line *no command "…" in the browser's core (drs-wasm 0.81.0): the app's core has it, this build not yet*.

Opening

item	does
New Plot Set…	an empty set, then Add Files… for its members (choose them in the tree first)
Open Plot Set…	opens the `.drv` chosen in the tree
Open Buffer…	opens the WinPLOTR `.buf` chosen in the tree
Open CSV as a Set…	the chosen files as one set (what Open as Plot Set does)
Add Files…	adds the files chosen in the tree to the set shown
Open Sequence (.seq)…	FullProf's sequential results against their external parameter **(later)**
Open Results Series…	a series of refinements against T **(later)**
Send Series to Publish…	**(app)**, always greyed here

The open panels of the browser are the tree: **choose the files in the tree first, then the command.** With nothing suitable chosen, the status line says what to choose.

Saving and output

item	does
Save Plot Set / Save Plot Set As…	downloads the `.drv` (§13)
Reload Members	reads the members' files again **(later)**
Plot Set as CSV…	wide or long CSV of the set **(later)**
Send to Publish…	**(app)**: saves (downloads), then says Publish is not in this app

Views and map tools

item	does
Show Overlay / Stack / Projection / Map / Difference Map / Surface / Series	switch the view (✓ on the one shown)
Copy Surface as Image	the surface as a PNG on the clipboard
Linked Zoom	the overlay's linked pair (✓)
Set Colour Range… / Reset Colour Range	type the clip / back to the core's range (map views and Surface)
Row and Column Cursors	the strips (✓)
Range from the View	the clip follows the zoom (✓)
Save Region… / Rename Region… / Remove Region	the map's regions
Y Interval Adapted	✓ = "y: own", off = "y: shared"
Viewer Page… (Dev)	the self-contained map page, downloaded

Choosing members: Choose All Members, Choose No Member, Choose by Exposure Class…, Choose by a Range of the Variable…, List Order…

Figure

item	does
Figure Look	the view as a figure (✓)

item	does
Set Scale Bar…	the scale bar's length in a stack or projection figure (empty: the core's round step)
Figure Size…	journal and thesis widths from Publish's presets **(later)**

Normalisation and x: Normalise…, Convert X…, Set Wavelength… (§12).

Order: Reverse Order, Sort by Variable, Read Variable From… (the file's name or the file itself work; *a temperatures file* is **(later)**).

Windows: Add Window…, Draw Window (✓), Rename Window…, Remove Window, Remove the Windows, Background at….

Opacity: Solo Member, Dim Others (others at 22 % opacity), Restore All.

Per member: Set Displacement…, Set Reference (also switches to the Difference Map), Set Temperature….

Derived: Sum of Members, Average of Members, Difference of Members (exactly two chosen: the first less the second). The core computes the pattern with its σ; the browser downloads it as `<set>sum.xys` (average, difference) and the status line says so.

Open in Data: **(app)**.

15. Live mode: a folder still being written

At a beamline or a diffractometer, scans keep arriving in a folder. In **Chromium**, with a folder opened by **Open Folder…** or dropped, the page can watch it:

- **Live** (checkbox, on by default): read the folder again on a timer;
- **every**: 2, 5, 10, 30 or 60 s (10 s by default);
- **Now**: read it at once.

Your choice is kept for the next visit. At each reading:

- **new files** appear in the tree and are described by the core;
- if files were added, the core **proposes again**, and every set **opened from a proposal** grows with the new scans the core now proposes for the same kind and class. They are added to the set as Add Files… would, and the map is computed again. Such sets are marked **live** in the proposals list;
- a file that **vanishes** is named in the live line (and in the status line if it belongs to a set), but it **stays in its set**: nothing is ever removed from a set by itself.

The live line under the controls says what is happening: "Reading the folder every 10 s; 1 live set. …".

Notes:

- Only sets opened **from a proposal** grow. A set opened from chosen files stays as it was opened.
- Re-proposing reads every data file again. For a hundred large scans this takes tens of seconds, which is why it is done only when files have been added.

- Browsers slow the timers of a tab in the background (to once a minute after five minutes). Keep the tab in front for quick updates.
- Firefox and Safari do not give the page a lasting handle on a folder, so live mode is not offered there.

16. The view in the address: links and reloads

The page keeps the view shown in its address, after the `#`, updating it about once a second:

- the folder's name;
- where the set came from: its files, its `.drv` or `.buf`, or the core's proposal by kind and class;
- the view;
- the colour scale;
- the map's zoom;
- the clip;
- the region chosen.

For example:

```
…/diff-ref-struct-web/#folder=1_Tb2pd2sn&set=buf%3ATb_vs_T.buf&view=map&scale=log&zoom=24%2C36%2C4%
```

The address holds **only names and numbers already on screen**, never a file's content. So:

- **Reload** the page, open the same folder again, and the view comes back as it was: the set opened, the view, the scale, the zoom, the clip and the region.
- **Send the link** to a colleague who has the same files: when they open their copy of the folder, they see the same view. If files are missing, the page names them and says to open that folder.
- **Bookmark** views you come back to.

The zoom is applied once the core's map has been drawn. You may see the whole map for a moment first.

Cite this view (Help ▾) copies one line with the tool's version, the core's, the date and this address, for a lab book or a paper's methods: *J. M. Igartua, diff-ref-struct Web 1.0.0 (core 0.81.0), view of 26 October 2026: https://…/#folder=….* With the same files, the line brings the same view back.

17. Offline and installed

On the first visit the browser keeps the page's own files: the page, the module, the core in WebAssembly, about 36 files and 4.4 MB. From the next visit on, the page opens **without the network**. The line under the inspector says what is kept: "Offline: 36 files, 4.4 MB kept by this browser (web …, core 0.81.0)". It never keeps your data files.

The guide offline too. On the same first visit the browser also keeps the guide's pages and this manual's PDF (not the screenshots, which it keeps once you have seen them), so the manual opens at a beamline without network.

Install the page as an app from the browser: in Chrome, the install icon in the address bar or ⋮ ▸ Cast, save and share ▸ Install page as app; in Safari, File ▸ Add to Dock; in Edge, ⋯ ▸ Apps ▸ Install. It then opens in its own window, from the Dock or the Start menu.

Updates. When a new version is published, the browser fetches it in the background and replaces the old one. A tab already open may keep running the old version until it is reloaded. If the version shown (start page, offline line) is older than expected, **close every tab of the page and open it again**.

18. Mouse, keyboard, appearance

On any plot

action	does
drag a box	zoom both axes
⇧-drag, middle-drag, or Space + drag	pan
two-finger scroll	pan (a mostly vertical scroll zooms x)
⌘-wheel or pinch	zoom about the pointer
wheel over an axis's margin (maps)	zoom that axis only
double-click	fit the whole plot
double-click on a colour bar	the core's colour range again
drag a colour bar's mark	clip the colour range
click (map with Cursors on)	pin or release the cursors
⌥-click a curve (Overlay)	solo it, and again to restore
drag on the plot with Draw window on	a new window
drag a window's edge	move it

In the tree and the members: click, ⌘-click (Ctrl-click), ⇧-click to choose; double-click to open (tree) or to set a value (member without one); ⌥-click to solo (members).

In the bottom table:

- click a header to sort (▲, ▼, off);
- drag a column's edge to resize it;
- ↑ ↓ move the selection;
- ⌘C copies the selected rows;
- **Find…** searches the table, and **Columns** shows or hides columns (time and title are hidden by default).

Elsewhere: Escape closes the Plots menu; ← → on the focused separator set the inspector's width.

Keyboard shortcuts. The Plots commands have no keyboard shortcuts in the browser: use the menu. In particular **⌘S is the browser's "Save page"**, not Save Plot Set. The status line's "Save (⌘S)" is the Mac app's wording.

Appearance: the ◐ button cycles auto, light and dark.

Accessibility: the tree, the menu, the separator and the status line carry their roles for screen readers.

The status line is announced politely as it changes.

19. Messages you will meet, and what to do

the status line says	meaning, and what to do
Nothing to open as a plot set: x.pcr (…)	the files chosen are not data files. Choose data files, a `.drv` or a `.buf`.
… choose the files in the folder's tree first (click, ⌘-click, ⇧-click), then the command again	a menu command needs files: choose them in the tree, then the command.
Plots: N members, k without a T (drawn apart) …	opened; k files gave no value of the variable. Set Temperature… places them.
no command "…" in the browser's core (drs-wasm 0.81.0): the app's core has it, this build not yet	the command needs the Mac app or a later web version (§20).
The core says: … (under a map)	the colour range typed hides much of the map; the core says how much.
To lies above from. / The pair … lies wholly outside the core's range	the clip typed is reversed, or outside the data.
Show the map or the difference map first: a region is the rectangle it shows.	switch to a map view before Save Region….
A region needs a name of its own … / A window needs a name of its own …	the name is taken.
Say where the value comes from: it is kept with it.	a typed temperature or λ needs its source.
Choose two members: the first chosen less the second.	Difference of Members needs exactly two chosen.
Downloaded as x.drv (your browser's Downloads): nothing is written into the folder …	Save worked: the file is in Downloads.
The address keeps a view of …'s files; n of them are not in the folder open …	the link or reload names files of another folder: open that one.
Live: … vanished from the folder; the set keeps it as read	a scan disappeared; the set keeps the data it read.
x.drv is open already: its tab brought forward.	the set was open; nothing new opened.
…: opened from its N files. The M values its set file types … are not carried	a proposal opened from its files; the variable was read again from the files (§7).

If the page seems stuck on a very large set, wait for the status line. A run of 101 synchrotron scans of 27 000 points (112 MB) takes about a minute to describe and about 40 s to propose. If nothing changes for several minutes, reload: the address brings the view back.

20. Limits of this version

What the browser does **not** do in diff-ref-struct Web 1.0.0 with core 0.81.0, and why. Each limit is on the roadmap (Appendix C) with what is decided about it.

- **Proposals against time open by temperature** (§7). A hold's or a run's members take the temperature in their names as their variable, and the time becomes the second variable. The map of a hold collapses. Use a `.drv` that types the times.
- **No figure export** (PDF, SVG, PNG): the Mac app's Export ▸ Plot. Use Figure look and a screenshot, or Viewer Page… for an interactive page.
- **No CSV export** (Plot Set as CSV…), no Figure Size…, no Reload Members, no Open Sequence (.seq)…, no Open Results Series…, no *temperatures file* in Read Variable From…: the browser's core does not serve these yet, and the status line says so.
- **No Data operations, refinement, Publish or projects**: those are the Mac app's.
- **Text files only**: binary formats (Bruker RAW and BRML, Rigaku RASX, NeXus/HDF5, zarr) are not read by the browser in this version, even though the core knows them (§21).
- **The strips under the map** (Cursors) may sit slightly off the map's frame in some window sizes: a peak in the row strip a degree or so left of its column, the strip's y label saying its scale twice ("counts (log) (log)"). It was seen on the public page on 26 October 2026; in other captures the frames match. Read 2θ on the map itself. A fix is under way.
- **Coming next**: a 2θ interval set once for the whole analysis (to cut away, for example, a beam-stop's first degree everywhere), and the map's default colour scale linear in place of log.

21. The file formats the core reads

The core recognises a file by its **content first**, its extension second, and says which reader it chose (hover the file in the tree). Text formats, read in the browser:

kind (as the tree shows it)	format	extensions
`xye`	x, y, σ columns (Topas .xye, GSAS-II's rules)	.xye .xy .dat .txt
`xys`	x, y, σ columns with six header lines (FullProf XYDATA)	.dat .xys .xy
`qye`	Q, y, σ columns, wavelength in a comment	.qye
`table`	table of columns: CSV, TSV, blanks; delimiter, decimal mark, header and columns recognised	.csv .tsv .txt .dat .xy
`free_steps`	FullProf free format: Ti, step, Tf, then the intensities	.dat
`ill_d1a_d2b_old`, `ill_d1b_old`, `ill_d1b_d20`, `ill_d2b`, `ill_d4_d20l`	ILL instruments (D1A, D2B, D1B, D20, D4)	.dat

kind (as the tree shows it)	format	extensions
`psi_dmc`	PSI DMC, HRPT	.dat
`brookhaven`	Brookhaven synchrotron (y and σ on alternate lines)	.dat
`two_axis`	general two-axis instrument	.dat
`variable_time`	X-rays with variable counting time	.dat
`dbws`	DBWS synchrotron format	.dat
`gsas`	GSAS powder data (STD, ESD, FXY, FXYE, ALT)	.gsas .gss .gsa .gda .fxye .fxy .raw .dat .xra .cwn
`isis_tof_banks`	ISIS TOF multibank	.gss .dat
`socabim_uxd`	Socabim UXD (Bruker DIFFRAC-AT text)	.uxd
`panalytical_xrdml`	PANalytical XRDML	.xrdml .xml
`rigaku_ras`, `rigaku_txt`	Rigaku RAS, Rigaku text export	.ras .txt
`fit2d_chi`	Fit2D or pyFAI .chi (2θ or Q)	.chi .qchi
`powder_cif`	powder CIF (pdCIF)	.cif

The project files it opens: plot sets (`.drv`) and WinPLOTR buffers (`.buf`).

Recognised by the core but **not read in the browser** (binary): Bruker RAW v1–v4, Bruker BRML, Rigaku RASX, NeXus/HDF5, MIDAS zarr. Use the Mac app or convert them to `.xye`.

Appendix A. Conventions

The words the page and its manuals use, the files it meets and what their suffixes mean, how to name scans so that the core reads them, the units and signs, and what the versions and ids mean.

Words

word	what it means
plot set	Several data files shown together; each is a member, with its value of the set's variable.
member	One pattern of a plot set: a file (or one scan of a file of several), its value of the variable and its source.
variable	What orders the members and places them on the map: a temperature, a time, or a value typed with its source.
source (of a value)	Where a value comes from: a file's header, its name, its title, a temperatures file, or a note typed with it. Every value keeps one.
second variable	Another value per member (the temperature of a scan in a set against time), shown on the map's right-hand axis.
proposal	A plot set the core finds in a folder: by temperature, a hold against time, the whole run against time.

word	what it means
hold	Three or more scans at the same temperature, followed against the time since the first of them.
exposure class	Scans counted alike, told apart by the core from the scale of their counts (the median of σ^2/y above 2°): long and short scans fall in different classes.
rebin	The points of a scan averaged onto a coarser regular grid (0.01°) by the core as the set opens; the files are untouched.
clip	The colour range chosen on the colour bar inside the core's full range; values beyond it are drawn saturated.
region	A named rectangle of a map (2θ and variable), with its own colour range, saved with the set.
window	A range of 2θ whose quantities (integral above the background, centroid, strongest peak) the Series view follows through the set.
reference member	The member subtracted in the Difference Map and in the Linked zoom's difference strip.
view rule	Normalise… or Convert X…: a change of how the set is drawn; the files are never changed.
live set	A set opened from a proposal that grows with the new scans live mode finds in the folder.
the core	diff-ref-struct's engine (Rust), here compiled to WebAssembly: it computes every number the page shows.

Files and suffixes

The project's own kinds come from its file dictionary (the same table that names the files the core writes); the others are files the page meets.

name or suffix	kind	what it holds	where
`{original}`	Measured data	The measured pattern(s) as the instrument wrote them, never changed.	`Data/`
`{data} processed.xys`	Data after the operations	The scan after the control file's data operations (masks, smoothing, rebinning, …): x, y, σ.	`Data/`
`{set} {operation}.xys`	Derived pattern of a plot set	The sum, average or difference of members of a plot set on a common grid, with σ, its '!' lines naming the members, each with its value of the series variable and the source of that value (Plots' PL9).	`Plots/`

name or suffix	kind	what it holds	where
`{set}.drv`	Plot set	Several data files as one set (Plots' PL12): the members (each file relative to the project's folder, a scan, a value of the series variable typed or read from a temperatures file, with its source (PL23), visible, opacity, colour), the variable's name and unit, the x quantity, the y scale, the order, the displacement rule, the reference member, the windows; the views' steps and scales, the normalisation (PL27); the map's and the difference map's colour range as clipped by the user (map_range, difference_range), and the regions of the variable–x plane saved by name ([[regions]]: x with the quantity it was drawn in, the variable's range, a colour range of its own, the window it came from) (PL37).	`Plots/`
`{name}.drp`	Publication set	What a document of the Publish module holds (R30, L094): the source (control file, version, results file), the sample's facts the files cannot give, and the figures, tables and CIFs with their options, built from those sources only (publish/DESIGN.md §5).	`Publish/`
`{project}.drc`	Control file	The refinement: the pattern and its data file, the phases and their atoms (or symmetry modes), what is refined, the cycles, the conventions. A joint refinement's (P06, L329): its patterns' control files (pattern_control) with their weights, refined flags and phases, the phases' names, and the parameters the patterns share ([[joint.pattern]], [[joint.phase]], [[joint.tie]]).	the project's folder
`{original}`	The app's state	What the app keeps about the project: the versions' metadata, the open windows, the last run (L067, L080).	`.drs/`
`.buf`	WinPLOTR buffer	A list of data files with a value each, written by FullProf's WinPLOTR; opens as a plot set, its files taken from its folder.	
`.xye, .xy, .dat, .xys, .qye, .chi, .gsas, .xrdml, .ras, .cif …`	Data files	Read by the core's readers by their content first, their extension second (Manual §21).	

name or suffix	kind	what it holds	where
`the address after #`	A view link	folder, set, view, scale, zoom, clip, region of the view shown: names and numbers, never a file's content (Manual §16).	

Naming scans

The core places a scan without help when its name says the temperature or the time. A sequence number first keeps the folder in acquisition order. Examples are file names the core reads as shown.

in a name	the core reads
`_300K, T300, 1p5K`	a temperature in K (1p5K is 1.5 K)
`T0900C, 900C, 900°C`	a temperature in °C; a set whose every value is in °C stays in °C
`t16530s`	the elapsed time in seconds (a lower-case t before the seconds)
`_long, _short`	words only: the core finds exposure classes from the counts, not from the name
`scan_001_T0900C_t00000s_long.xye`	sequence, temperature, time, exposure: sortable and read by the core

Units and signs

quantity	unit
2θ	degrees
d	Å
Q = 4π sin θ / λ, sin θ / λ, 1/d	$Å^{-1}$
intensity	counts as read; √counts or $\log_{10}$ counts on those colour scales
a window's integral	counts·° above the background, with σ
temperature	K, or °C when every value of the set is in °C
time	s, min or h since the set's first scan, as the set says
numbers	a decimal point

Versions and ids

form	what it is
`diff-ref-struct Web 1.0.0`	the tool's version (semantic: major.minor.patch); its release notes say what each brought
`core 0.81.0`	the version of the core compiled into the page; every number is that core's
`build a1c308d`	the commit the page was built from
`#12`	a request (bug or feature) in the public repository's issues

form	what it is
`L429, I10, W11`	the project's log entries, ideas and the web stream's requests, cited by the release notes and the roadmap

Appendix B. Release notes

Every version of diff-ref-struct Web up to 1.0.0, the one this manual describes, newest first.

1.0.0 · The guide, open source, the record

2026-10-27 · published · core 0.81.0 · build b4820d8

The page gains its own site, a licence and a citation, a place for requests, and a record (releases, log, roadmap, conventions) that the site and the manuals render; versions become semantic.

Added

- The guide at /guide/: overview, a five-minute start with a check of what the browser supports, the manual and a one-page quick reference (HTML and PDF), five recipes, how it works, formats, conventions, releases, log, roadmap, requests, about.
- The manuals as PDF, typeset by LuaLaTeX in the site's typefaces; every edition with three appendices: Conventions, Release notes, Roadmap.
- The guide offline: after a first visit the guide's pages and the manual's PDFs open without network, as the page does.
- A Help menu in the page: the guide, the manual at the section of the view shown, what's new, Report a problem…, Ask for a feature…, Cite this view.
- Try the example: a synthetic heating run computed by the core (21 scans, 300 to 1300 K, a second phase appearing), opened as if a folder had been dropped.
- Open source: the code under the Apache License 2.0, the guide, manuals and screenshots under CC BY 4.0; CITATION.cff; issue forms for bugs and features.
- Semantic versions: diff-ref-struct Web 1.0.0, with the core's version and the build beside it.

Fixed

- The Overlay draws its curves: the browser's core named each pattern's member `index` where the page reads `member`; the page reads both (a rename, no number touched).
- Sum, Average and Difference of Members download their file (they were computed and lost).

Known

- Proposals against time open with each scan's temperature as its variable: open a hold or a run from a .drv that types the times (recipe: A hold).
- No figure or CSV export in the browser; text formats only.

Closes W11; log L431, L433, L434; evidence: Web host tests 40 of 40 with the core 0.81.0 package; every guide screenshot taken from the app in both themes; the public PDFs and the guide checked for team

data by the leak guard.; archived as doi:10.5281/zenodo.23682727.

0.4.0 · The strips take the map's frame

2026-10-26 · published · core 0.81.0 · build a1c308d

The Plots module's step 23: the row strip under the map takes the map's frame, so that a peak sits under its column; the scale's word said once.

Known

- The Overlay draws no curves (the patterns' member named otherwise by the browser's core); the Stack and the maps are not affected.
- On the public page, in some window sizes, the strips may still sit off the map's frame.

log L431, L432, L433.

0.3.0 · Public at jmigartua.github.io/diff-ref-struct-web

2026-10-25 · published · core 0.81.0 · build 24d8619

The first publication by its Action, after the page's steps 4 and 5.

Added

- Live mode (Chromium): the folder read again every 2 to 60 s, new files described, sets opened from a proposal grown with the new scans.
- Offline: the page keeps its 36 files (4.4 MB) and opens without the network; installable as an app.
- The view in the address: the set, view, colour scale, zoom, clip and region kept in the link.
- Series recognised by the core: one scan per temperature, the holds, the run against time; exposure classes from the counts' scale σ^2/y; large scans rebinned to 0.01° on opening.
- A hundred raw synchrotron scans opened and overlaid in seconds: the patterns asked when drawn, the overlay drawn in slices.

log L427, L428, L429.

0.2.0 · The core in the browser: files described, buffers and set files routed

2026-10-25 · not published · core 0.79.0 · build bf451a3

Added

- The core compiled to WebAssembly answers every command; each file described by the core (its kind in the tree, listed by folders or by kinds).
- A .drv or a .buf among the files opens as itself, its members taken from the folder.

log L424, L425, L427.

0.1.0 · The browser host

2026-10-25 · not published · core — · build —

Added

- The app's Plots module loaded as it is, in a page: the inspector, the folder's tree, the separator, the views.

log L423.

Appendix C. Roadmap

What is decided, considered and set aside after this version.

Decided

- **Proposals against time opened as their set files.** A hold or a run then opens at its times, as the core proposed it; today it opens by temperature. (W10)
- **The core answers in the app's shapes, with a test of the shapes.** So that a renamed field never again breaks a view in the browser (the Overlay, 0.4.0). (W11)
- **A 2θ interval set once for the whole analysis.** Every view, window and export then sees the cut set: a beamstop's first degree left out everywhere. (L434, PL42)
- **The map's colour scale linear by default.** Josu's ask; a set file that says its scale keeps it. (L434)
- **The cursor strips on the map's frame in every window size.** Seen off by a degree or two on the public page. (L433)
- **The inspector's panels opening their own manual section.** The view's ? does it from 1.0.0; the members, regions and windows panels need a hook in the Plots module. (W12)
- **Screenshots as a release check.** A view drawn blank stops the release; the blank Overlay of 0.4.0 was found this way. (web/site/tools/shots.mjs –check)

Considered

- **Spanish and Basque editions of the guide and the manuals.** Asked for later; English only for now. Basque through the EHU rules (EHULKU), with the decimal comma.
- **Figure export in the browser (PDF, SVG, PNG).** The Mac app's Export ▸ Plot; in the browser a screenshot of Figure look serves a draft.
- **CSV export, sequences (.seq) and results series in the browser.** The core has them; its WebAssembly build does not serve them yet.
- **Binary formats in the browser (Bruker RAW and BRML, Rigaku RASX, HDF5).** The page reads files as text; reading bytes needs the host and the core's package to pass them.
- **The page's interface translated.** After the manuals; the shell has no translation layer yet.

Set aside

- **Hosting FullProf's WinPLOTR tutorial files as the example.** They are the ILL's to distribute; the example the core computes needs no permission, and the tutorial stays linked as the second example.
- **A request form on a third-party service.** It would send users' words to a server we do not control; the issues are public and need no data.
- **The whole private repository made public.** It holds other programs' manuals and a team's unpublished data; the public repository takes the subset the page is built from, by an Action.